LECTURE 1.2 · SHIP AND OCEAN SYSTEMS DESIGN

Owner’s Requirements

Papanikolaou (2026) §1.3.1, §1.3.5; §1.3.6.1–1.3.6.2 for Activity 1.2 · Session W1.2, first half · Lecture 20 min + Activity 1.2 20 min

Tiago Santos, Luís Batista

Instituto Superior Técnico / University of Lisbon

tiago.santos@centec.tecnico.ulisboa.pt

luis@veranavis.com

calendly.com/vnsd-luisbatista/meetme

The Holy Bible, Genesis 6–8: Owner’s Requirements to Noah

Verse	The owner’s words (King James Version, 1611)	As a design requirement
6:14	Make thee an ark of gopher wood; rooms shalt thou make in the ark, and shalt pitch it within and without with pitch.	Timber hull, divided into compartments, sealed with pitch inside and outside
6:15	The length of the ark shall be three hundred cubits, the breadth of it fifty cubits, and the height of it thirty cubits.	L 135 m · B 22.5 m · D 13.5 m, with 1 cubit ≈ 0.45 m; L/B = 6, L/D = 10
7:20	Fifteen cubits upward did the waters prevail; and the mountains were covered.	Draught not stated; one reading: T = 15 cubits ≈ 6.75 m, half of D (an interpretation to max draught)
6:16	A window shalt thou make to the ark … and the door of the ark shalt thou set in the side thereof; with lower, second, and third stories shalt thou make it.	Light and air opening under the roof; one side door; three decks
6:18	… thou shalt come into the ark, thou, and thy sons, and thy wife, and thy sons' wives with thee.	8 persons on board, who are also the crew
6:19	… two of every sort shalt thou bring into the ark, to keep them alive with thee; they shall be male and female.	Cargo: live animals in pairs, many species: pens, feeding, handling
6:21	And take thou unto thee of all food that is eaten … and it shall be for food for thee, and for them.	Food for people and animals for the whole voyage; no resupply
7:12, 7:24	And the rain was upon the earth forty days and forty nights. … And the waters prevailed upon the earth an hundred and fifty days.	Watertight against 40 days of rain and months afloat
7:11, 8:14	In the six hundredth year of Noah's life, in the second month, the seventeenth day … And in the second month, on the seven and twentieth day … was the earth dried.	Endurance: just over one year (about 370 days), no port calls
7:18, 8:4	… the ark went upon the face of the waters. … And the ark rested … upon the mountains of Ararat.	No propulsion, no steering: it drifts, and must survive grounding

Potential solutions can be found in: https://www.fulltable.com/VTS/n/nk/na.htm

Who Buys the Ship and sets the Requirements?

Who buys and owns the ship	Contract	Who crews and maintains it	Who chooses cargo and route	Big names (top 5: next slide)
Shipping line liner: fixed routes and schedules	**Owner-operator** no charter: own ships	Shipping line, or its ship manager	Shipping line	**MSC, Maersk, CMA CGM**, the largest (2026): each owns part of its fleet and charters the rest
Tonnage provider owns ships and charters them out to shipping lines	**Time charter** hire per day	Tonnage provider, or its ship manager	Shipping line (charterer)	**Seaspan, Danaos, Costamare, Global Ship Lease**: container ships built to charter out
Tramp shipowner no fixed route: goes where the cargo is	**Voyage charter** freight per voyage	Tramp Shipowner	Charterer, for one voyage	Tanker and bulk owners: **COSCO, Frontline, Star Bulk**
Shipowner keeps ownership; rents the ship to an operator	**Bareboat charter** the ship alone, without crew, for years	Charterer crews and runs it as “disponent owner”	Charterer	Common in leases: the lessor’s ship, bareboat-chartered to a shipping line
Lessor bank or leasing house; holds the title	**Sale and leaseback** operator sells the ship, rents it back	Lessee the operator that uses the ship	Lessee	Chinese leasing houses: **Bocom, ICBC, CMB Financial Leasing**

Charterer: hires a ship from its owner and decides how it is used: for one voyage (pays freight), for a period with the owner’s crew (pays hire per day), or bare, with its own crew (as disponent owner). Usually a shipping line, a trader, an oil major or a mining company.

Buyer
signs the yard contract

Ship manager
runs ships for owners

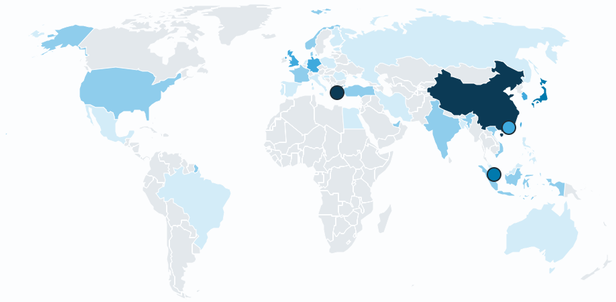
Beneficial owner
controls the owning firm

Hire / freight
per day / per voyage

Extended reading: **Reading 1.2 Who Buys the Ship: Real Cases**, one real deal per row

Big Names: the Top 5 in Each Role

Shipping lines containers, million TEU		Tonnage providers containers, million TEU		Shipowners, all types million dwt		Ship managers third-party, by ships managed
1 MSC	7.34	**1** Seaspan	1.95	**1** COSCO	76.2	**1** Anglo-Eastern
2 Maersk	4.72	**2** Shoei Kisen	0.84	**2** MSC	44.5	**2** OSM Thome
3 CMA CGM	4.38	**3** Eastern Pacific	0.63	**3 Bocom Leasing**	41.4	**3** V.Group
4 COSCO	3.64	**4** Zodiac	0.61	**4–5** China Merchants		**4** Synergy Marine
5 Hapag-Lloyd	3.40	**5** Costamare	0.52	**4–5** MOL		**5** Bernhard Schulte
Alphaliner, July 2026. Hapag-Lloyd includes ZIM, bought in 2026		Alphaliner, April 2025		MSI ranking; lessor in blue; 10th: ICBC Leasing, 19.9		Lloyd’s List Top 10, 2025



Who controls the world fleet · beneficial owner, million dwt, 2026

1 Greece	397		**6** South Korea	99.2	
2 China	377		**7** Germany	69.0	
3 Japan	243		**8** Taiwan	66.2	
4 Singapore	182		**9** United Kingdom	61.1	
5 Hong Kong	122.8		**10** UAE	59.6	

Extended reading: **Reading 1.2B Who Owns the World Fleet**, country by country

From a Business Need to the Owner’s Requirements

1 · The need

Why owners order new ships

- Replace old or non-compliant ships
- Expand a service already run
- Start a service on a new route
- Carry a new cargo
- Adopt a new technology or fuel
- Special ships, offshore units

Our two ships: replace an ageing fleet on the Azores lines

The owner’s decision

How the need becomes a mission

1. Study the market and the fleet it needs
2. Compare: move, charter, buy second-hand or build new
3. If new: write the mission

Owner’s engineering writes the brief and specification

The mission: type, area, capacity, speed, range, crew, propulsion

2 · The requirements

What the owner specifies

a Capacity	deadweight, volume, TEU, lanes, passengers
b Speed	trial speed at the design draught
c Range	miles or days without bunkering
d Class	the classification society
Often added	propellers, engine type (not its power), cargo handling, accommodation

Left to the designer: main dimensions, except canal and port limits; stability

Economic interests: a contract is signed only when they cross: the owner’s return, the yard’s price and slot, the bank’s risk and the market’s demand.

Contractual requirements: price and payment schedule, delivery and delay penalties, guarantees. Often set by lenders or the future charter, they share the risk between yard and owner.

ACTIVITY 1.2 · IN PAIRS · 20 MIN · NOT GRADED · PRACTICE FOR D1

Does the Owner’s Brief Add Up?

With your neighbour, on **the reference ship’s brief** (0.6, slide 2). Round numbers and a phone calculator are enough. Fuel at sea about **25 t/day**; water, stores and crew about **300 t**.

1 Volume · 4 min

Does 35,000 t of cargo at 0.90 t/m³ fit in the cargo tanks at 98 % filling? Hands up: yes or no.

2 Deadweight · 6 min

37,600 t minus 35,000 t of cargo leaves 2,600 t. Enough for fuel for 10,000 nm at 14.5 kn (about 29 days), plus water, stores and crew?

3 Other cargoes · 5 min

A light product at 0.75 t/m³ and a heavy chemical at 1.20 t/m³: for each, what runs out first, the tanks or the deadweight? Guess, then check.

4 Classify · 2 min

A deadweight carrier or a volume carrier (Ship types 1.2)? Class vote.

Together · 3 min: for each step one pair gives its answer, and the class checks it.

Your project: your group checks its own ship’s brief the same way outside class, for D1 (see your Project Scope Sheet).

Four Things to Take Away

The buyer writes the brief. Whoever signs the contract, a line, a tonnage provider or a lessor, decides what the ship must be.

A ship is an investment. It is built only when it pays better than moving, chartering or buying one.

The owner sets the what. Capacity, speed, range and class: values that can be checked at delivery.

The designer finds the how. Main dimensions, power and form, but only once the brief adds up.

Next, in this session: Lecture 1.3, The Design Process. Keep your numbers from Activity 1.2.